WHEN ARE TRIANGLES INVISIBLE? FUNDAMENTAL LIMITS OF HIGHER-ORDER STRUCTURE IDENTIFIABILITY FROM EDGE FLOWS

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ABSTRACT

Topological signal processing models higher-order interactions by placing signals on the triangles of a simplicial complex, but in many applications the set of "filled" triangles is unknown and must be inferred from edge-flow observations. Existing work provides algorithms for this inference; we instead ask when it is *possible at all*. We show that the latent higher-order structure is observed only through the *curl* of the edge flows, which annihilates the gradient and harmonic parts of the signal. This reduces per-triangle detection to a two-variance Gaussian test whose optimal error exponent is a Gaussian Chernoff information, and yields a sharp *curl-invisibility* phase: when the curl signal-to-noise ratio ρ falls below $\rho^\star(N) = \Theta(\sqrt{\log(1/\delta)/N})$, no estimator can recover the structure from N snapshots. When candidate triangles share edges, a geometry-aware whitening of the curl statistic yields exact per-triangle marginal detection laws with effective SNR $\rho_\tau^{\mathrm{eff}} = \sigma_c^2/(\sigma_n^2(\boldsymbol{G}^{-1})_{\tau\tau})$ governed by the triangle Gram matrix $\boldsymbol{G}$, together with a rigorous union-bound recovery guarantee. Finally we expose a second, SNR-independent obstruction: supports are identifiable only modulo $\ker \boldsymbol{B}_2$, and on the complete graph K_n the observable curl subspace has dimension $\binom{n-1}{2}$ — a $3/n$ degrees-of-freedom ratio relative to the $\binom{n}{3}$ candidate triangle coefficients. All theorems are validated against Monte-Carlo simulation and illustrated on real data from both sides of the limit: arbitrage-free foreign-exchange flows, where the theory predicts (and we measure) that both obstructions bind, and road-network traffic flows, whose favorable planar geometry admits recovery at exactly the predicted snapshot budget. Code and data are released for full reproducibility.

Index Terms— Topological signal processing, simplicial complexes, Hodge decomposition, identifiability, detection theory, edge flows.

1. INTRODUCTION

Graph signal processing captures pairwise structure, but many networked measurements—currency exchange, traffic circulation, neural flow, statistical ranking—carry *higher-order* structure that lives on the triangles of a simplicial complex [1, 2, 3]. Topological signal processing (TSP) has rapidly built a toolbox on the Hodge Laplacian: convolutional filters [4], maximally localized dictionaries [5], Dirac operators [6], adaptive estimation [7], and unrolled reconstruction [8]. A recurring primitive is that the set of filled triangles is unknown and must be learned from edge-flow signals.

Prior work. Learning a graph topology from node signals is by now well charted [9, 10]. Its higher-order analogue — inferring which triangles of a complex are filled — has so far been treated *algorithmically*: sparse-clique-sampling MAP estimation [11], greedy simplicial topology learning [12], and sparse cell-complex augmentation [13] all return an estimate of the filled set, and adaptive schemes [7] list structure discovery as open. HodgeRank [14] established the curl of an edge flow as the signature of higher-order inconsistency (e.g. arbitrage). What is missing is the *converse side* of the problem: none of these works characterizes when the task is information-theoretically solvable, at what snapshot budget, and how the answer depends on the geometry of the complex. Detectability thresholds for simplicial/hypergraph stochastic block models observe the higher-order structure *directly*; here the structure is latent and observed only through the flows it shapes. This paper supplies the missing fundamental limits, using classical detection and information-theoretic tools [15, 16]. Our contributions are:

1. A *curl-annihilation* reduction (Lemma 1): the latent structure is observable only through the flow's curl, turning per-triangle detection into a two-variance Gaussian test whose optimal error exponent is a Gaussian Chernoff information (Prop. 1), and making support recovery over edge-disjoint candidates obey an *exact* finite-sample product law (Prop. 2).
2. A sharp *curl-invisibility* phase transition (Theorem 1): the structure is unrecoverable whenever ρ falls below $\rho^\star(N) = \Theta(\sqrt{\log(1/\delta)/N})$, with $C(\rho) \sim \rho^2/16$.
3. A *geometry-aware* whitened detector for edge-sharing (confusable) candidates with exact per-triangle marginal laws and effective SNR ρ_τ^{eff} (Theorem 2), a rigorous union-bound recovery guarantee (Cor. 1), and empirical dominance over naive curl-energy thresholding.
4. An SNR-independent *rank obstruction* (Theorem 3): supports are identifiable only modulo $\ker \boldsymbol{B}_2$; on K_n the observable curl subspace carries only a $3/n$ degrees-of-freedom ratio.

Every result is validated against Monte-Carlo simulation, and the limits are illustrated on real data from both sides: foreign-exchange flows (both obstructions bind; nothing is recoverable) and road-network traffic flows (favorable geometry; planted structure is recovered at the predicted budget).

2. BACKGROUND AND SIGNAL MODEL

2.1. Hodge structure and notation

Fix a graph $(\mathcal{V}, \mathcal{E})$ oriented so that the node–edge incidence is $\boldsymbol{B}_1 \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{E}|}$ and, for a chosen candidate set $\mathcal{T}$ of triangles (e.g. all 3-cliques), the edge–triangle incidence is $\boldsymbol{B}_2 \in \mathbb{R}^{|\mathcal{E}|\times|\mathcal{T}|}$. Throughout, N denotes the number of observed snapshots and $p = |\mathcal{T}|$ the

Code: `github.com/appleweiping/topo-flow-limits`.

number of candidate triangles. The boundary-of-boundary identity $\boldsymbol{B}_1\boldsymbol{B}_2 = \mathbf{0}$ holds, and the Hodge 1-Laplacian $\boldsymbol{L}_1 = \boldsymbol{B}_1^\top \boldsymbol{B}_1 + \boldsymbol{B}_2\boldsymbol{B}_2^\top$ induces the orthogonal decomposition

$$\mathbb{R}^{|\mathcal{E}|} = \underbrace{\operatorname{im} \boldsymbol{B}_1^\top}_{\text{gradient}} \oplus \underbrace{\operatorname{im} \boldsymbol{B}_2}_{\text{curl}} \oplus \underbrace{\ker \boldsymbol{L}_1}_{\text{harmonic}}. \tag{1}$$

The *curl* of an edge flow is $\operatorname{curl} \boldsymbol{f} = \boldsymbol{B}_2^\top \boldsymbol{f}$; a flow induced by a triangle signal $\boldsymbol{y}$ is $\boldsymbol{B}_2\boldsymbol{y}$. Each column of $\boldsymbol{B}_2$ has exactly three ± 1 entries, so the triangle Gram matrix $\boldsymbol{G} = \boldsymbol{B}_2^\top \boldsymbol{B}_2$ has $G_{\tau\tau} = 3$ and $G_{\sigma\tau} \in \{0, \pm 1\}$, with $G_{\sigma\tau} \neq 0$ iff σ and τ share an edge.

2.2. Generative model

The 1-skeleton is known; the latent object is the set $\mathcal{S} \subseteq \mathcal{T}$ of filled triangles. We observe N i.i.d. edge-flow snapshots

$$\boldsymbol{f}_t = \boldsymbol{B}_1^\top \boldsymbol{a}_t + \boldsymbol{B}_{2,\mathcal{S}}\, \boldsymbol{y}_t + \boldsymbol{h}_t + \boldsymbol{n}_t, \quad t = 1, \dots, N, \tag{2}$$

with node potentials $\boldsymbol{a}_t \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \boldsymbol{I})$ (gradient nuisance), triangle signals $\boldsymbol{y}_t \sim \mathcal{N}(\mathbf{0}, \sigma_c^2 \boldsymbol{I})$ supported on $\mathcal{S}$, harmonic nuisance $\boldsymbol{h}_t \in \ker \boldsymbol{L}_1$, and noise $\boldsymbol{n}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \boldsymbol{I})$; $\boldsymbol{B}_{2,\mathcal{S}}$ keeps the columns of the active triangles. The goal is to recover $\mathcal{S}$ from $\{\boldsymbol{f}_t\}_{t=1}^N$. This is the generative family used by the algorithmic literature (cf. [11]); we characterize when its inverse problem is solvable.

3. FUNDAMENTAL LIMITS

3.1. Curl annihilation and the single-triangle test

Lemma 1 (Curl annihilation)**.** *For any $\boldsymbol{f}$, the statistic $\boldsymbol{B}_2^\top \boldsymbol{f}$ depends only on the curl component: $\boldsymbol{B}_2^\top(\boldsymbol{B}_1^\top \boldsymbol{a}) = \mathbf{0}$ for all $\boldsymbol{a}$ and $\boldsymbol{B}_2^\top \boldsymbol{h} = \mathbf{0}$ for all $\boldsymbol{h} \in \ker \boldsymbol{L}_1$. Under (2),*

$$\boldsymbol{c}_t \;=\; \boldsymbol{B}_2^\top \boldsymbol{f}_t \;=\; (\boldsymbol{B}_2^\top \boldsymbol{B}_{2,\mathcal{S}})\, \boldsymbol{y}_t + \boldsymbol{B}_2^\top \boldsymbol{n}_t. \tag{3}$$

Proof. $\boldsymbol{B}_1\boldsymbol{B}_2 = \mathbf{0}$ gives $\boldsymbol{B}_2^\top \boldsymbol{B}_1^\top = \mathbf{0}$. If $\boldsymbol{L}_1\boldsymbol{h} = \mathbf{0}$ then $0 = \boldsymbol{h}^\top \boldsymbol{L}_1 \boldsymbol{h} \geq \boldsymbol{h}^\top \boldsymbol{B}_2 \boldsymbol{B}_2^\top \boldsymbol{h} = \|\boldsymbol{B}_2^\top \boldsymbol{h}\|^2$, so $\boldsymbol{B}_2^\top \boldsymbol{h} = \mathbf{0}$. ☐

Thus the gradient and harmonic nuisances—however large—are removed exactly by the curl map, and only the active triangles and projected noise remain. All information about $\mathcal{S}$ flows through (3).

For an *isolated* candidate τ (sharing no edge with other candidates) the scalar $c_{\tau,t}$ is zero-mean Gaussian with variance $v_0 = 3\sigma_n^2$ if $\tau \notin \mathcal{S}$ and $v_1 = 9\sigma_c^2 + 3\sigma_n^2 = v_0(1+\rho)$ if $\tau \in \mathcal{S}$, where

$$\rho \;\triangleq\; 3\sigma_c^2/\sigma_n^2 \tag{4}$$

is the *curl signal-to-noise ratio*.

Proposition 1 (Two-variance detection)**.** *Detecting an isolated τ from N snapshots is a two-variance Gaussian test: the energy $E = \sum_{t=1}^N c_{\tau,t}^2$ is sufficient, $E/v_i \sim \chi_N^2$ under H_i, the minimum Bayes error is attained by thresholding E and admits a closed form via χ_N^2 tails, and it decays as $\exp(-N\, C(v_0, v_1))$, where $C(v_0, v_1) = \max_{s \in [0,1]} g(s)$ is the Gaussian Chernoff information [15, 16], with $w_i = 1/v_i$ and*

$$g(s) = \tfrac{1}{2}\big[\log(sw_0 + (1-s)w_1) - s\log w_0 - (1-s)\log w_1\big]. \tag{5}$$

Theorem 1 (Curl-invisibility phase)**.** *With ρ as in (4), $C(\rho) \to 0$ as $\rho \to 0$ with $C(\rho) = \rho^2/16 + o(\rho^2)$. Consequently, for target error δ and budget N, reliable detection requires $\rho \geq \rho^\star(N) = \Theta\big(\sqrt{\log(1/\delta)/N}\big)$; below $\rho^\star$ the triangle is unrecoverable by* any *estimator.*

Proof sketch. The Chernoff information is the optimal Bayes error exponent for i.i.d. observations, so no test beats $\exp(-NC)$; Taylor-expanding $C(v_0, v_0(1+\rho))$ at $\rho = 0$ gives $\rho^2/16$, whence $\exp(-NC(\rho)) \leq \delta$ forces $\rho = \Omega(\sqrt{\log(1/\delta)/N})$. ☐

3.2. Support recovery: the edge-disjoint case

When the candidates in $\mathcal{T}$ are pairwise edge-disjoint, the coordinates of $\boldsymbol{c}_t$ are *independent* across triangles (the columns of $\boldsymbol{B}_2$ have disjoint supports and the noise is white), so support recovery decouples exactly.

Proposition 2 (Exact finite-sample recovery law)**.** *For edge-disjoint candidates and the per-triangle Bayes-optimal energy threshold, the probability of exact support recovery equals $\prod_{\tau \in \mathcal{S}}(1 - P_{\mathrm{miss}}) \prod_{\tau \notin \mathcal{S}}(1 - P_{\mathrm{fa}})$, with $P_{\mathrm{miss}}, P_{\mathrm{fa}}$ given by χ_N^2 tails at the common threshold. Its 50% contour in the (N, ρ) plane is the empirical phase boundary (Fig. 1).*

3.3. Confusable triangles: geometry-aware whitening

Edge-sharing candidates are *confusable*: an active triangle leaks curl energy onto neighbours through the off-diagonal entries of $\boldsymbol{G}$, so naive curl-energy thresholding fails — and *worsens* as ρ grows, since stronger signals leak harder (Fig. 2, left). Whitening repairs this.

Theorem 2 (Whitened marginal laws)**.** *Assume $\boldsymbol{B}_2$ has full column rank, so $\boldsymbol{G}$ is invertible. Let $\hat{\boldsymbol{y}}_t = \boldsymbol{G}^{-1}\boldsymbol{c}_t$ and let $\tilde{\boldsymbol{y}}_t$ denote $\boldsymbol{y}_t$ zero-padded to all of $\mathcal{T}$. Then, exactly,*

$$\hat{\boldsymbol{y}}_t = \tilde{\boldsymbol{y}}_t + \boldsymbol{e}_t, \qquad \boldsymbol{e}_t \sim \mathcal{N}\big(\mathbf{0},\, \sigma_n^2 \boldsymbol{G}^{-1}\big). \tag{6}$$

Hence the marginal *law of each coordinate $\hat{y}_{\tau,t}$ is a two-variance test with $v_{0,\tau} = \sigma_n^2(\boldsymbol{G}^{-1})_{\tau\tau}$, $v_{1,\tau} = \sigma_c^2 + v_{0,\tau}$, and per-triangle effective SNR*

$$\rho_\tau^{\mathrm{eff}} = \frac{\sigma_c^2}{\sigma_n^2(\boldsymbol{G}^{-1})_{\tau\tau}}. \tag{7}$$

Proof. By Lemma 1, $\boldsymbol{c}_t = \boldsymbol{G}\tilde{\boldsymbol{y}}_t + \boldsymbol{B}_2^\top \boldsymbol{n}_t$ (the columns of $\boldsymbol{G}$ indexed by $\mathcal{S}$ act on $\boldsymbol{y}_t$; zero-padding absorbs the indexing). Applying $\boldsymbol{G}^{-1}$ gives the mean exactly; the noise covariance is $\sigma_n^2 \boldsymbol{G}^{-1}\boldsymbol{G}\boldsymbol{G}^{-1} = \sigma_n^2 \boldsymbol{G}^{-1}$. Gaussian marginals of (6) give the per-coordinate two-variance laws. ☐

The factor $(\boldsymbol{G}^{-1})_{\tau\tau}$ is the price of geometry: it equals $1/3$ for an isolated triangle (recovering $\rho_\tau^{\mathrm{eff}} = \rho$ of (4)) and grows as curl signatures overlap. The coordinates of $\boldsymbol{e}_t$ are correlated whenever $\boldsymbol{G}^{-1}$ is not diagonal, so the joint recovery probability does *not* factorize in general; what survives arbitrary correlations is a union bound over the exact marginals:

Corollary 1 (Rigorous recovery guarantee)**.** *Threshold each $\sum_t \hat{y}_{\tau,t}^2$ at its marginal-optimal (or Bonferroni) level and let $P_{\mathrm{err},\tau}$ be the resulting marginal error probabilities from Theorem 2. Then $\Pr[\hat{\mathcal{S}} = \mathcal{S}] \geq 1 - \sum_{\tau \in \mathcal{T}} P_{\mathrm{err},\tau}$, regardless of the correlation structure of $\boldsymbol{e}_t$.*

Remark 1 (Independence approximation)**.** *The product $\prod_\tau(1 - P_{\mathrm{err},\tau})$ is exact whenever the whitened scores are independent — for edge-disjoint candidates (Prop. 2), or more generally whenever $\boldsymbol{G}$ is diagonal — and is an approximation otherwise. In our benchmarks $\boldsymbol{G}$ is diagonally dominant, the off-diagonal correlations of $\boldsymbol{G}^{-1}$ are weak, and the product law, the union bound, and Monte-Carlo coincide within 0.025 (Fig. 2, right).*

3.4. The rank obstruction

Whitening cannot help when curl signatures are linearly dependent, i.e. when $\boldsymbol{G}$ is singular.

Theorem 3 (Geometry-side degrees-of-freedom obstruction)**.** *If* $\operatorname{im} \boldsymbol{B}_{2,\mathcal{S}} = \operatorname{im} \boldsymbol{B}_{2,\mathcal{S}'}$ *then* $\mathcal{S}$ *and* $\mathcal{S}'$ *induce identical flow distributions and are indistinguishable at any SNR and any* N*: without further priors, supports are identifiable only modulo* $\ker \boldsymbol{B}_2$*. On the complete graph* K_n *with all* $\binom{n}{3}$ *triangles as candidates,* $\dim \operatorname{im} \boldsymbol{B}_2 = \binom{n-1}{2}$*, so the ratio of observable curl degrees of freedom to candidate triangle coefficients is exactly* $3/n$*.*

Proof. Only $\operatorname{im} \boldsymbol{B}_{2,\mathcal{S}}$ enters the distribution of (2), giving the indistinguishability claim. For K_n the 2-skeleton is simply connected, so $\operatorname{im} \boldsymbol{B}_2 = \ker \boldsymbol{B}_1$, the cycle space, of dimension $|\mathcal{E}| - (n-1) = \binom{n-1}{2}$; dividing by $\binom{n}{3}$ gives $3/n$. $\square$

Remark 2 (Sparsity can restore identifiability)**.** *Theorem 3 concerns* arbitrary *supports. Under a sparsity prior, specific supports with* $|\mathcal{S}|$ *small may remain identifiable even when* $p > \operatorname{rank} \boldsymbol{B}_2$*, provided the participating columns satisfy compressed-sensing-type (spark/coherence) conditions; characterizing those sparse thresholds is an open problem beyond this paper's scope.*

4. EXPERIMENTS

All experiments run on CPU from fixed seeds; every theoretical curve is overlaid on independent Monte-Carlo simulation, and a released test-suite checks each closed form against simulation. Code and data reproduce all figures via a single script.

4.1. Phase transition (validation of Thm. 1, Prop. 2)

Fig. 1 sweeps (N, ρ) on an edge-disjoint planted complex ($p = 8$, 4 active) with gradient and harmonic nuisance present. The empirical exact-recovery probability undergoes a sharp transition; the solid curve is the exact product law of Prop. 2, whose empirical 50% contour it tracks to a median ratio of 1.01 across the grid. The dotted curve is the asymptotic floor $\rho^\star \sim 1/\sqrt{N}$ of Theorem 1.

4.2. Confusability (validation of Thm. 2, Cor. 1)

On an edge-sharing triangle strip ($p = 9$, alternating 4 active, $\rho = 8$; $\boldsymbol{B}_2$ has full column rank), the naive curl-energy detector plateaus at a mean Hamming error of ≈ 2.6 and does not improve with N, while the whitened detector of Theorem 2 recovers the support exactly (Fig. 2, left). Its empirical exact-recovery probability lies between the rigorous union bound (Cor. 1) and the independence approximation (Remark 1), all three agreeing within 0.025 (Fig. 2, right). A greedy residual-fitting baseline (standing in for greedy topology learning [12]) stalls at ≈ 0.3.

4.3. FX flows: both obstructions bind (converse side)

We form edge flows from 257 trading days of ECB daily reference rates over nine currencies (complete graph K_9; Fig. 3). This study *illustrates the obstructions* — it is not a recovery benchmark, since real FX carries no ground-truth triangle support. (A) An arbitrage-free market is a pure gradient flow [14]: measured curl energy is 10^{-31} of gradient energy — machine zero — so higher-order structure is invisible at any budget, exactly as Theorem 1 predicts for

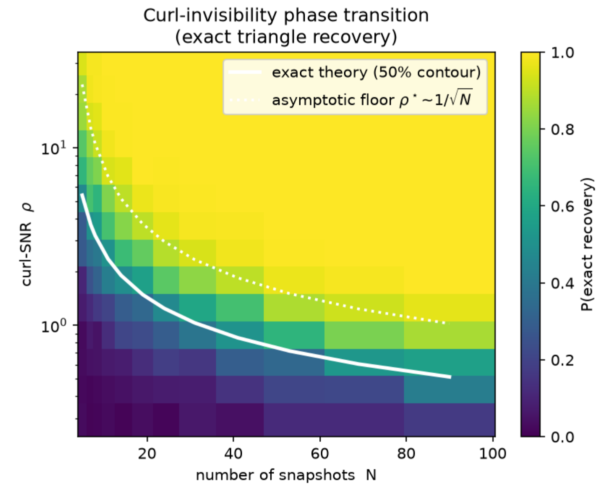

Fig. 1. Curl-invisibility phase transition for exact triangle recovery. Colour is empirical $P(\text{exact recovery})$; the solid line is the exact finite-sample law (Prop. 2), the dotted line the $1/\sqrt{N}$ asymptotic floor (Thm. 1).

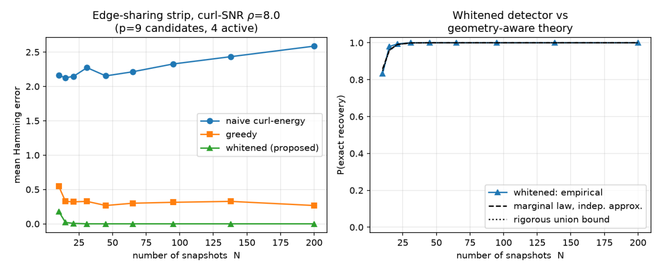

Fig. 2. Edge-sharing strip. Left: mean Hamming error vs. N; the naive detector never recovers. Right: whitened detector vs. the exact marginal theory — independence approximation (dashed) and rigorous union bound (dotted).

$\rho \to 0$. (B) The measured curl rank on K_9 is 28 against 84 candidate coefficients, matching the $3/n$ degrees-of-freedom law of Theorem 3 (verified numerically for K_5–K_{12}).

4.4. Road traffic: favorable geometry and achievability

Road networks sit on the opposite side of the limit (Fig. 4). (A) On three standard traffic-assignment networks (TNTP: Sioux Falls, Eastern-Massachusetts, Anaheim [17]), the planar-like street geometry contains few 3-cliques and $\boldsymbol{B}_2$ has *full column rank* in all three — curl degrees-of-freedom ratio exactly 1, against $3/n$ on K_n: every candidate triangle is identifiable. Unlike FX, the real user-equilibrium flows carry genuine rotational energy (2.4% curl on Sioux Falls, 4.6% on Anaheim). (B) Achievability at the predicted budget: on Anaheim ($p = 54$ candidate triangles, pairwise edge-disjoint, so $\boldsymbol{G} = 3\boldsymbol{I}$ and the product law is exact), we plant 18 active triangles on top of the *real* equilibrium flow (plus Gaussian gradient/harmonic nuisance) at $\rho = 3$ and run the whitened detector on temporally centered scores — centering removes any constant background flow exactly at the cost of one degree of freedom, so the theory curves use $N-1$. Empirical recovery matches the exact

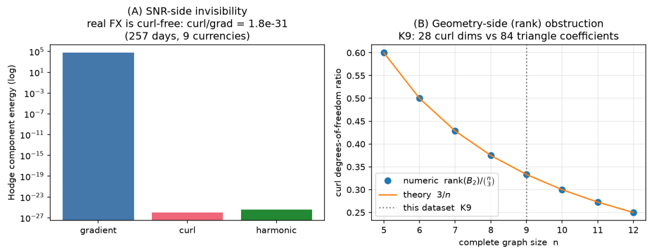

Fig. 3. Real FX flows (illustration). (A) SNR-side invisibility: curl energy is machine-zero relative to gradient. (B) Geometry-side obstruction: curl degrees-of-freedom ratio equals $3/n$.

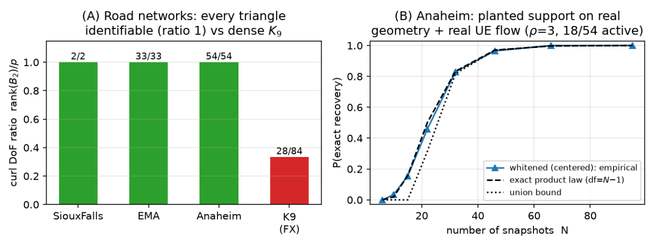

Fig. 4. Road-network traffic. (A) All three TNTP networks have full-column-rank $\boldsymbol{B}_2$: DoF ratio 1 (green) vs. K_9's $3/n$ (red). (B) Planted-support recovery on Anaheim's real geometry over the real equilibrium flow: empirical matches the exact df-$(N-1)$ product law.

product law across the whole transition. Planted signals are used because real traffic carries no labelled triangle support; what is real is the geometry and the equilibrium-flow background the detector must be invariant to. Together with the FX study this closes the loop: the same theory that certifies impossibility on K_9 predicts—quantitatively—when recovery succeeds on real road geometry.

5. CONCLUSION

We characterized when higher-order structure can be identified from edge flows: a curl-invisibility phase transition in the curl-SNR, exact per-triangle marginal laws with a geometry-aware effective SNR after whitening, a rigorous union-bound recovery guarantee, and a rank obstruction that no SNR can overcome. The results delimit when algorithmic TSP inference [11, 12, 13] can succeed, and real data instantiate both sides: FX flows where the obstructions bind, road networks where recovery succeeds at the predicted budget. Open directions include sub-Gaussian converses, identifiability under partial edge sampling [18], plug-in estimation of the variance parameters, and sharp sparse-support thresholds (Remark 2).

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